



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

MISINFORMATION DETECTION

USING NLP AND MACHINE LEARNING

The domain of the Project:

Artificial Intelligence & Machine Learning

Team Mentors (and their designation):

Mr. Gaurav Patel
Artificial Intelligence & Machine Learning Trainer

Team Members:

Sameera Appikonda

Period of the project

April 2026 to June 2026

Abstract

Misinformation can reduce public trust and make it difficult to distinguish reliable news from misleading or fabricated content. This project presents an end-to-end Misinformation Detection system that analyzes news text and predicts whether it is likely to be fake or real. The system is designed as a practical AI/ML workflow that connects data preparation, natural language processing, feature extraction, supervised learning, evaluation and an interactive application.

The implementation combines text preprocessing, TF-IDF feature extraction and Logistic Regression classification. The Streamlit application presents the prediction together with probability scores, confidence visualization, key-word analysis and an illustrative spread simulation. The project therefore demonstrates not only model training but also how a trained model can be exposed through a usable interface.

Keywords

Misinformation Detection • Fake News • NLP • TF-IDF • Logistic Regression • Machine Learning • Streamlit • ISOT Dataset

Table of Contents

1. Introduction	Page 3
2. Problem Statement	Page 3
3. Objectives	Page 3
4. Project Architecture	Page 4
5. Dataset and Data Processing	Page 5
6. Text Preprocessing Pipeline	Page 5
7. Feature Engineering	Page 6
8. Machine Learning Model	Page 6
9. Exploratory Data Analysis	Page 7
10. Deployment and Application	Page 8
11. Project Output / Demonstration	Page 9
12. Testing and Expected Behavior	Page 10
13. Limitations and Future Scope	Page 10
14. Conclusion	Page 10
15. References	Page 10
16. Community Services	Page 11

1. Introduction

The rapid growth of online information has made it increasingly difficult to distinguish reliable news from misleading or fabricated content. Automated text classification can assist users by identifying linguistic patterns associated with fake news. The objective of this project is to build a machine-learning-based text classification system capable of identifying news content as fake or real.

The implementation follows an end-to-end pipeline rather than relying only on a classifier. It includes dataset preparation, normalization and cleaning, feature extraction, model development, evaluation, visualization and an application interface. This structure makes it possible to move from raw news records to a prediction that can be presented to a user through a dashboard.

2. Problem Statement

Design and implement an automated system that accepts a news headline or article as text, processes the content using NLP techniques, represents the text numerically, and predicts whether the news is likely to be fake or real. The system should also communicate the prediction clearly and provide supporting probability information so that the output is understandable to a user.

3. Objectives

- Clean and normalize raw news text.
- Remove URLs and unnecessary non-alphabetic characters.
- Remove English stopwords using NLTK.
- Convert textual information into numerical features using TF-IDF.
- Train a Logistic Regression classifier for fake/real classification.
- Evaluate the model using accuracy, classification metrics and ROC-AUC.
- Generate visualizations for dataset and model analysis.
- Provide an interactive Streamlit application for real-time prediction.
- Display probability scores and supporting analysis for the prediction.

4. Project Architecture

The project is organized as a sequence of connected stages. Each stage prepares information for the next stage, creating a complete pipeline from user input and training data to a prediction shown in the Streamlit application.

Stage	Description
Input	Raw news headline or article text supplied by the user.
Data Processing	Loads Fake.csv and True.csv, assigns labels and combines title and text.
Preprocessing	Lowercase conversion, URL removal, non-alphabetic character removal and stopword removal.
Feature Engineering	TF-IDF representation of the cleaned text with a maximum of 5,000 features.
Modeling	Logistic Regression classification for fake/real prediction.
Evaluation	Accuracy, precision, recall, F1-score, confusion matrix and ROC/AUC analysis.
Deployment	Streamlit application using the trained model and supporting visualizations.

Project Directory Structure

MISINFORMATION_DETECTOR/

- data/
- models/
- notebooks/
- src/
- app/
- plots/
- eda.py
- requirements.txt

Deployment

The trained model and preprocessing components are used by the Streamlit application to perform prediction on user-entered news text. The application combines the classifier output with probability displays and supporting visual elements so that the result is easier to interpret.

5. Dataset and Data Processing

The project uses the ISOT Fake News Dataset. The application dashboard reports 23,502 fake articles and 21,417 real articles, giving a total of 44,919 articles. The records are prepared by assigning class labels and combining the title and article text before the NLP preprocessing stage.

Dataset Detail	Value
Dataset	ISOT Fake News Dataset (UVic)
Fake articles	23,502
Real articles	21,417
Total articles	44,919
Training/Test split	80% / 20%
Random state	42

6. Text Preprocessing Pipeline

The preprocessing stage converts raw news content into a consistent form suitable for machine learning. The same general preparation is applied during training and when the application receives new input.

- Combine the news title and article text.
- Convert text to lowercase.
- Remove URLs.
- Remove non-alphabetic characters.
- Remove English stopwords using NLTK.
- Handle missing records before model training.
- Pass cleaned text to TF-IDF feature extraction.

This preprocessing produces a consistent text representation for both model training and application-time prediction. Cleaning reduces unnecessary variation in the input while retaining the words used by the classifier to distinguish the two classes.

7. Feature Engineering

TF-IDF (Term Frequency–Inverse Document Frequency) transforms cleaned news text into numerical features. The vectorizer is configured with a maximum of 5,000 features. This representation allows the Logistic Regression classifier to work with the textual patterns extracted from the dataset.

How TF-IDF is used

The vectorizer converts the cleaned collection of news documents into a numerical matrix. Frequently occurring words that are less useful for distinguishing documents receive lower importance, while terms that are informative within the document collection can receive higher weights. The resulting feature matrix is passed directly to the classifier as part of the training pipeline.

8. Machine Learning Model

The project uses Logistic Regression for binary classification. The model works with the TF-IDF representation and produces a fake/real decision together with probability scores in the application.

Model Configuration	Value
Algorithm	Logistic Regression
Feature representation	TF-IDF
Maximum features	5,000
Training/Test split	80% / 20%
Reported accuracy	98.88%
Reported AUC score	0.99

Evaluation

The evaluation stage considers classification performance using accuracy and ROC-AUC, with supporting visual analysis in the application. The reported dashboard values are 98.88% accuracy and an AUC score of 0.99 for the supplied project output.

9. Exploratory Data Analysis

The project includes analysis and visualization to understand the dataset and model behaviour. The implementation contains plotting code and the application also provides word-level and prediction-related visualizations. These views help connect the numerical model output with patterns visible in the data and submitted text.

Analysis Areas

- Label/class distribution.
- Text and article characteristics.
- Model classification performance.
- Confusion-matrix analysis.
- ROC/AUC evaluation.
- Important or frequently occurring words in the submitted input.

Model Information Shown in the Application

Item	Value
Algorithm	Logistic Regression
Features	TF-IDF (5,000)
Accuracy	98.88%
AUC Score	0.99
NLP	Stopword Removal

Interpretation of the Analysis

The analysis stage supports model development by making the class distribution and classification behaviour easier to inspect. The application extends this idea to an individual prediction by showing probability scores and a key-word view for the submitted news statement.

10. Deployment and Application

The project provides an interactive Streamlit dashboard named **MisInfo Shield**. It accepts a news statement and presents a classification result with probability scores and additional analysis. The interface connects the saved machine-learning pipeline with user input so that a prediction can be obtained without running the training process again.

Application Features

- News statement input.
- Fake/real prediction.
- Fake and real probability scores.
- Confidence breakdown visualization.
- Misinformation spread simulation.
- Key words in the submitted input.
- Dataset and model information.
- Model limitation and verification disclaimer.

Technology Stack

Technology	Role in the project
Python	Core programming language.
Pandas / NumPy	Data handling and numerical processing.
NLTK	Text preprocessing and stopwords removal.
Scikit-learn	TF-IDF and Logistic Regression.
Streamlit	Interactive web application.
Plotly	Interactive visualization.
Matplotlib / Seaborn	Project analysis and plots.

The deployment stage demonstrates the practical use of the trained model: a user enters news text, the application applies the same preprocessing and feature transformation, and the classifier returns the predicted class and probability information.

11. Project Output / Demonstration

The following screenshots show the actual MisInfo Shield application output supplied for this project.

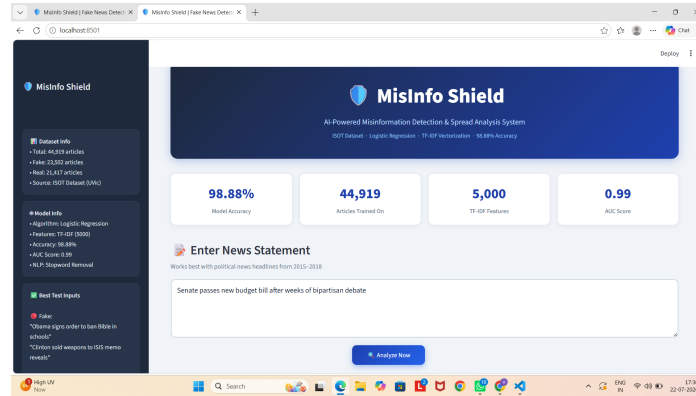


Figure 1. MisInfo Shield dashboard showing dataset and model information.

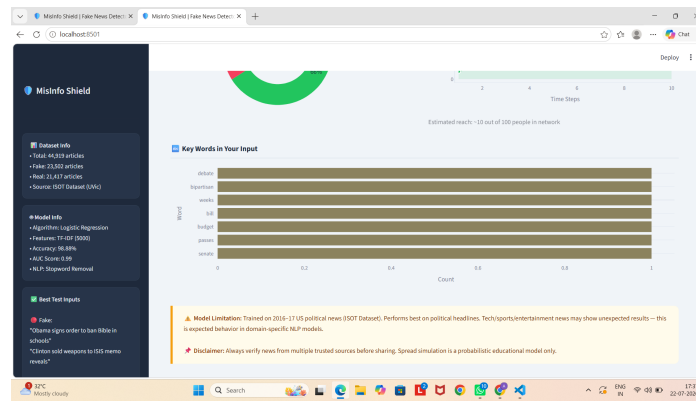


Figure 2. Analysis result showing REAL NEWS with 66.0% real probability and 34.0% fake probability.

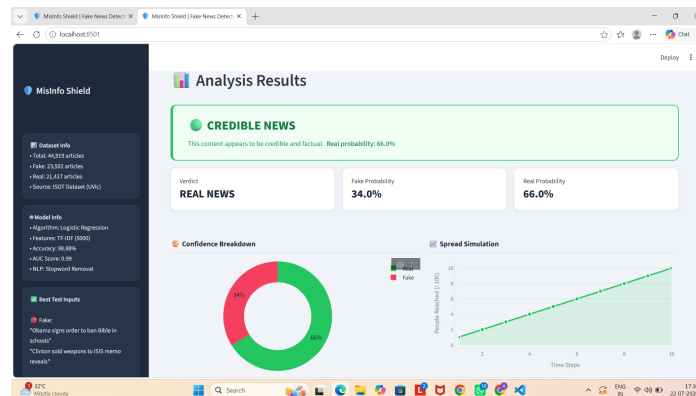


Figure 3. Key words, spread simulation and model limitation/disclaimer.

Observed Output

Item	Observed value
Input example	Senate passes new budget bill after weeks of bipartisan debate
Prediction	REAL NEWS
Real probability	66.0%
Fake probability	34.0%

12. Testing and Expected Behavior

Test case	Expected behavior
Valid news statement	System preprocesses the text and returns a fake/real classification.
Empty input	Application should request meaningful input before analysis.
Text containing URLs	URLs are removed during preprocessing.
Punctuation / numbers	Cleaning removes non-alphabetic characters.
Prediction request	Model returns a class prediction with probability scores.

13. Limitations and Future Scope

Limitations

- The model is trained on the ISOT dataset and may perform best on news patterns similar to its training data.
- Model probability is confidence from the classifier and is not absolute proof that a news item is true or false.
- News topics and writing patterns change over time, so periodic retraining may be required.
- The spread simulation is an illustrative educational model, not a measurement of actual social-media reach.
- The application recommends verifying news from multiple trusted sources before sharing.

Future Scope

- Use transformer-based models such as BERT or RoBERTa.
- Add multilingual misinformation detection.
- Integrate real-time news sources.
- Add explainable AI showing important words or feature contributions.
- Provide API/browser-extension support for wider use.
- Add model monitoring and drift detection.

14. Conclusion

The MisInfo Shield project demonstrates a complete applied AI/ML workflow for misinformation detection. It combines dataset preparation, NLP preprocessing, TF-IDF feature extraction, Logistic Regression classification, evaluation, visualization and Streamlit deployment. The supplied application demonstrates an end-to-end prediction experience with probability scores and supporting analysis.

15. References

- ISOT Fake News Dataset — University of Victoria (UVic).
- Project implementation and application components supplied for the MisInfo Shield project.
- Scikit-learn documentation — TF-IDF and Logistic Regression concepts.
- NLTK documentation — text preprocessing and stopword removal.
- Streamlit documentation — interactive machine-learning application development.
- Plotly documentation — interactive visualization.

16. Community Services

During the internship period, I participated in community-oriented activities in **Visakhapatnam, Andhra Pradesh**. These activities complemented the technical learning by providing opportunities to work with others, coordinate tasks and contribute to community welfare.

Activities Involved

Activity	Participation
Tree Plantation Drive	Participated in tree plantation activities in Visakhapatnam.
Food Distribution Activity	Participated in food distribution activities in Visakhapatnam.

Contribution

- Contributed towards environmental awareness and greener surroundings.
- Supported community welfare through food distribution.
- Improved teamwork, coordination, communication and social-responsibility skills.

Course and Trainer Details

Detail	Information
Course Opted	6-Month Project-Based Internship in Artificial Intelligence & Machine Learning
Instructor Name	Gaurav Sir
Duration	6 Months
Trainer Name	Gaurav Patel
Trainer Email ID	gaurav.patel.gpp@gmail.com
Trainer Designation	Data Science Instructor

Community Service Repository

<https://github.com/sameeraappikondag37python-cyber/SAMEERA-APPIKONDA-Community-Service>

Learning Outcomes

- Applied an end-to-end NLP and machine-learning workflow to a practical misinformation detection problem.
- Gained experience with preprocessing, TF-IDF, Logistic Regression, evaluation and Streamlit deployment.
- Strengthened communication, coordination, teamwork and social-responsibility skills through community activities.

Acknowledgments

I sincerely thank Prof. Radhakumari Challa, Executive Director and Founder of SURE Trust, Mr. Gaurav Patel, and SURE Trust for the internship learning opportunity and guidance.

— End of Project Report —